

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

UNIT I SIGNALS AND SYSTEMS

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: IT6502 Digital Signal Processing

Name of the Faculty member: Mrs.S.Jeeva

Name of the Topic: Introduction to Digital Signal Processing

Name of the Innovative Practice: Real time Examples

Description:

Digital signal processing (DSP) is the use of digital processing, such as by computers or more specialized digital signal processors, to perform a wide variety of signal processing operations.

DSP applications include audio and speech processing, sonar, radar and other sensor array processing, spectral density estimation, statistical signal processing, digital image processing, data compression, video coding, audio coding, image compression, signal processing for telecommunications, control systems, biomedical engineering, and seismology.

✚ **Audio world** : MP3 compression, Audio signal processing, Speech compression and transmission in digital mobile phones, Audio compression, hi-fi, Speech processing, speech Recognition.

✚ **Video** : Digital image processing, Video compression,

✚ **Others** : Digital communications, radar, sonar, Specific examples are, seismic data processing, analysis and control of industrial processes, medical imaging such as CAT scans and MRI, computer graphics, image manipulation.

✓ Real time Examples: Winamp, Digital Clock, OFDM, Human EEG & ECG, Fetal heart rate detection during labour.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1		1		1		1		1

CO1: The students will be able to analyze the properties of discrete time signals and properties of systems using Z-transform.

References:

1. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
2. Andreas Antoniou, 'Digital Signal Processing', Tata McGraw Hill, 2006.
3. S.K. Mitra, 'Digital Signal Processing – A Computer Based Approach', McGrawHill Edu, 2013.